End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B. Tech, CSE-AI, Semester:<VI>
(April -May, 2023)

Subject Code: BAI 306	Subject: Digital Image Processing (DIP)
Time :3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 **(2.5*8=20)**

(a) Differentiate between sampling and quantization.
(b) Describe various types of adjacency.
(c) Differentiate between low pass filters and high pass filters.
(d) Find Haar transform of the sequence –

$$f(m,n)=\begin{bmatrix}4 & -1\\ 2 & 3\end{bmatrix}$$

(e) Define Image segmentation. What is referred as a good segmentation?
(f) Differentiate between lossy and lossless compression.
(g) Define a structuring element.
(h) Define and explain Image Recognition.

UNIT-I

Q2 Given below are the two histograms (i) and (ii). Perform histogram matching and modify histogram (i) as given by histogram (ii). **(10)**

(i)

r_k	0	1	2	3	4	5	6	7
n_k	80	100	90	60	30	20	10	0

(ii)

r_k	0	1	2	3	4	5	6	7
n_k	0	0	0	60	80	100	80	70

Q3 Define and explain point operations in digital image processing. **(10)**
Consider the following given image:

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

(i) Perform Thresholding operation and draw it's histogram. (T=4)
(j) Perform Intensity Level Slicing (with background) for r_1=3 and r_2=5.

UNIT-II

Q4 Define noise and mention it's sources. Explain four different types of noise along with their PDF. **(10)**

Q5 Convert the given domain image using Fourier transform and perform IHPF to smoothen the image. Choose D_0 as 0.5. Show step by step procedure. **(10)**

1	0	1	0
1	0	1	0
1	0	1	0
1	0	1	0

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UNIT-III

Q6 What is region based segmentation? How is it different from Thresholding segmentation? **(10)**

Apply region growing on the following image with initial point at (2,2) and threshold value as 2. Use 4 connectivity.

0	1	2	0
2	5	6	1
1	4	7	3
0	2	5	1

Q7 Define Huffman Coding in image compression. **(10)**

Construct Huffman coding for each gray level and find compression ratio and coding efficiency.

Gray level	0	1	2	3	4	5	6	7
Freq.	500	200	250	100	250	500	150	50

UNIT-IV

Q8 Differentiate between dilation and erosion. **(10)**

Compute A⊕B for the following input image-

0	0	0	0	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

Q9 Explain object recognition and all the steps involved in object recognition. What are the central issues that should be considered in designing an object recognition system? **(10)**